Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	6 September 2026
Team ID	Individual
Project Name	Automated Incident Management Lifecycle using ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The proposed solution uses ServiceNow as the application platform. End users report IT issues through the Incident module in Service Operations Workspace. The incident is classified and supported using Agent Assist and Knowledge Management, then reassigned to the appropriate Level 2 support group. Level 2 technicians investigate the incident, create emergency changes or child incidents when required, document cause and resolution, and resolve the incident. Knowledge Articles and SLA tracking provide reusable knowledge and service visibility.

Architecture Flow

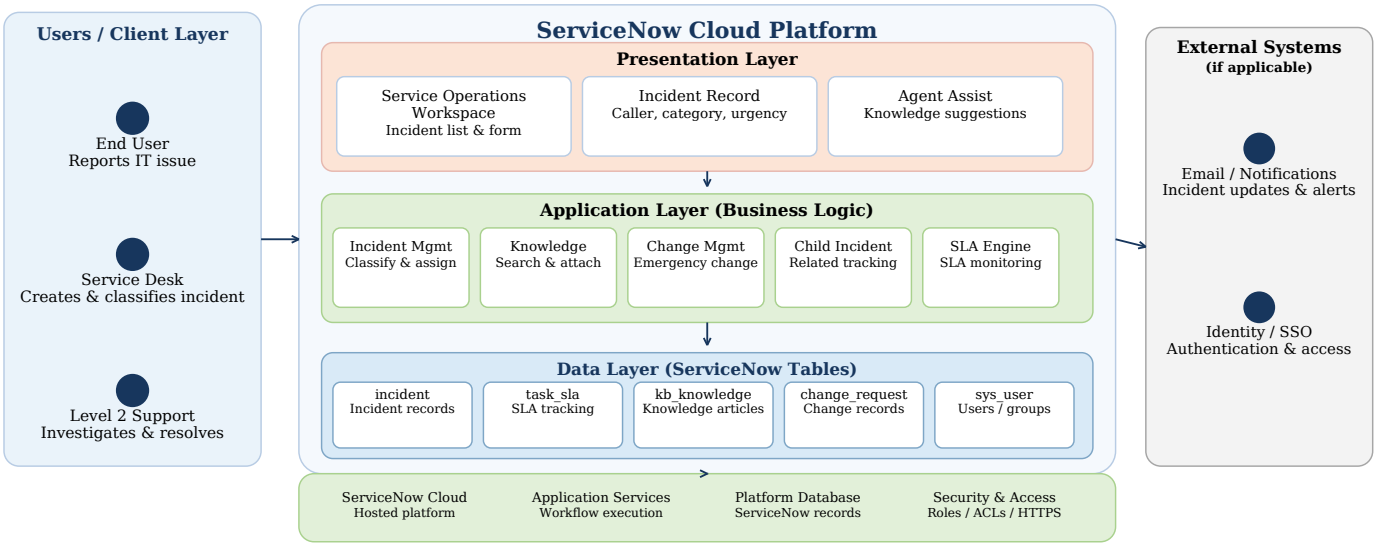


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which Service Desk agents and support teams create, classify, assign, and track incidents.	ServiceNow Service Operations Workspace
2	Application Logic-1	Captures incident details and manages classification, assignment, and incident lifecycle states.	ServiceNow Incident Management
3	Application Logic-2	Provides relevant troubleshooting knowledge and allows articles to be attached to incidents.	Agent Assist / Knowledge Management
4	Application Logic-3	Routes incidents to Level 2 groups and supports escalation and related-record workflows.	ServiceNow Assignment / Flow Designer
5	Database	Stores incidents, users, SLAs, knowledge, changes, and related task information.	ServiceNow tables: incident, sys_user, task_sla, kb_knowledge, change_request
6	Cloud Database	Platform-managed data storage for the ServiceNow application and related records.	ServiceNow platform data store
7	File Storage	No separate file-storage component is required for the core incident workflow.	Not applicable
8	External API-1	No external API is required for the core incident lifecycle.	Not applicable
9	External API-2	No external API is required for the core incident lifecycle.	Not applicable
10	Machine Learning Model	No machine-learning model is required for the proposed ServiceNow workflow.	Not applicable
11	Infrastructure (Server / Cloud)	Application and workflow are hosted and executed on the ServiceNow instance.	ServiceNow Personal Developer Instance / Enterprise Instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required. The solution is configured within the ServiceNow platform.	ServiceNow platform configuration
2	Security	Access to incidents, knowledge, changes, and related records is controlled using ServiceNow roles and permissions.	ServiceNow roles, ACLs, authentication and authorization
3	Scalable Architecture	The solution is platform-based and can support additional support groups, knowledge articles, workflows, users, and incident volume as usage grows.	ServiceNow platform and Flow Designer
4	Availability	The application runs on the ServiceNow platform, allowing Service Desk and support teams to access the incident workflow through the configured instance.	ServiceNow instance
5	Performance	The workflow reduces manual coordination by automating classification support, knowledge access, reassignment, change linkage, and related-record tracking.	Incident Management, Agent Assist and Flow Designer